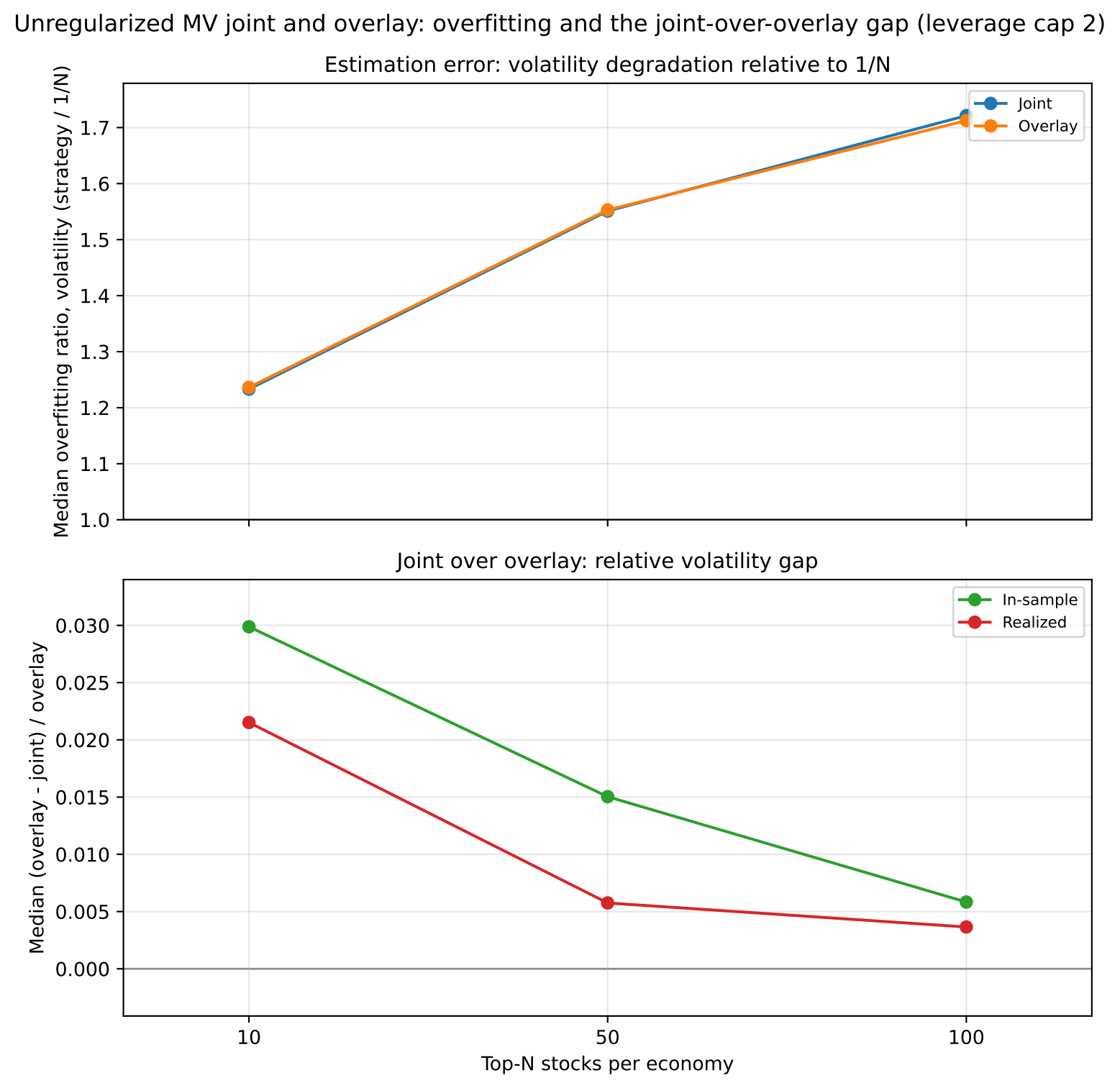
Unregularized MV joint and overlay: overfitting and the joint-over-overlay gap (leverage cap 2)
Estimation error: volatility degradation relative to 1/N
Median overfitting ratio, volatility (strategy / 1/N)
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
Joint
Overlay
Joint over overlay: relative volatility gap
Median (overlay - joint) / overlay
0.000
0.005
0.010
0.015
0.020
0.025
0.030
In-sample
Realized
10
50
100
Top-N stocks per economy